

1.1 Physical Quantities & Units

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation

$$v = k \left(\frac{\Delta P}{\rho} \right)^n$$

where k is a constant that has no units.

What is the value of n ?

- A. $\frac{1}{2}$
- B. 1
- C. $\frac{3}{2}$
- D. 2

[1 mark]

Question 2

Which estimate is realistic?

- A. The kinetic energy of a bus travelling on an expressway is 30,000 J
- B. The power of a domestic light is 300 W
- C. The temperature of a hot oven is 300 K
- D. The volume of air in a car tyre is 0.03 m^3

[1 mark]

Question 3

Which statement is **incorrect** by a factor of 100 or more?

- A. Atmospheric pressure is about $1 \times 10^5 \text{ Pa}$
- B. Light takes $5 \times 10^2 \text{ s}$ to reach us from the Sun
- C. The frequency of ultraviolet light is $3 \times 10^{12} \text{ Hz}$
- D. The life-span of a man is about $2 \times 10^9 \text{ s}$

[1 mark]

Question 4

The frictional force F on a sphere falling through a fluid is given by the formula

$$F = 6\pi a\eta v$$

where a is the radius of the sphere, η is a constant relating to the fluid and v is the velocity of the sphere.

What are the units of η ?

- A. kg m s^{-1}
- B. $\text{kg m}^{-1} \text{s}^{-1}$
- C. kg m s^{-3}
- D. $\text{kg m}^3 \text{s}^{-3}$

[1 mark]

Question 5

When a constant braking force is applied to a vehicle moving at speed v , the distance d moved by the vehicle as it comes to rest is given by the expression

$$d = k v^2$$

where k is a constant.

When d is measured in metres and v is measured in metres per second, the constant has a value of k_1 .

What is the value of the constant when the distance is measured in metres, and the speed is measured in kilometres per hour?

- A. $0.0772 k_1$
- B. $0.278 k_1$
- C. $3.60 k_1$
- D. $13.0 k_1$

[1 mark]

Question 6

The drag coefficient C_d is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where F is the drag force on the car, ρ is the density of the air, A is the cross-sectional area of the car and v is the speed of the car.

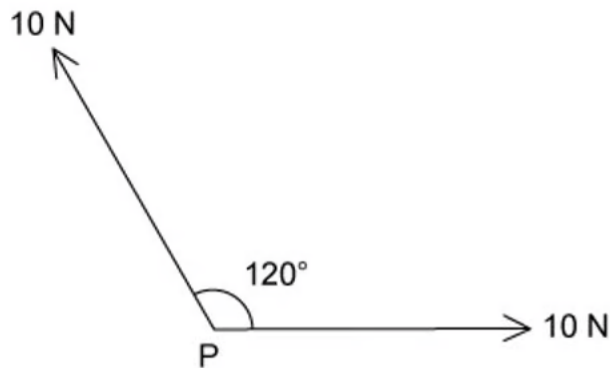
What is the value of n ?

- A. 1
- B. 2
- C. 3
- D. 4

[1 mark]

Question 7

Two forces, each of 10 N, act at a point P as shown in the diagram. The angle between the force vectors is 120°



What is the magnitude of the resultant force?

- A. 5 N
- B. 10 N
- C. 17 N
- D. 20 N

[1 mark]

Question 8

Which formula could be correct for the speed v of ocean waves in terms of the density ρ of sea-water, the acceleration of free fall g , the height h of the wave and the wavelength λ ?

A. $v = \sqrt{\rho gh}$

B. $v = \sqrt{\frac{g}{h}}$

C. $v = \sqrt{g\lambda}$

D. $v = \sqrt{\frac{g}{\rho}}$

[1 mark]

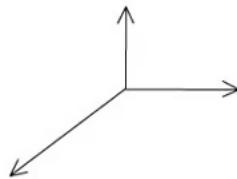
Question 9

The diagrams represent systems of coplanar forces acting at a point. The lengths of the force vectors represent the magnitudes of the forces.

Which system of forces is in equilibrium?

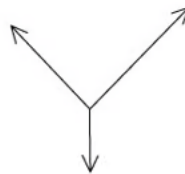
A.

A



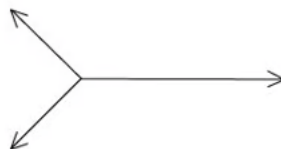
B.

B



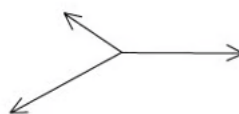
C.

C



D.

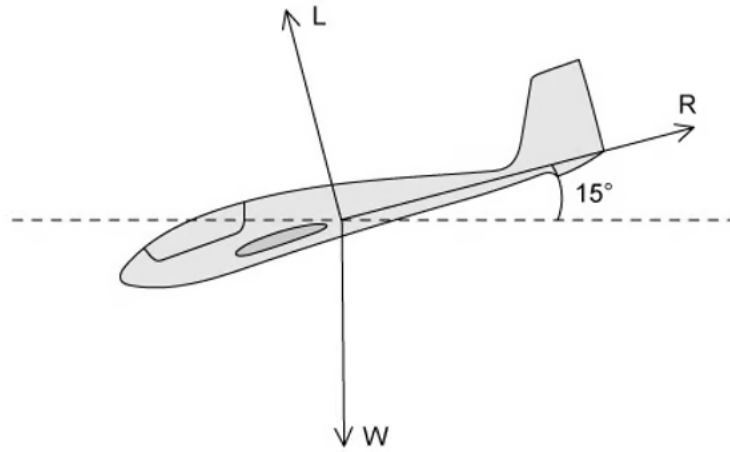
D



[1 mark]

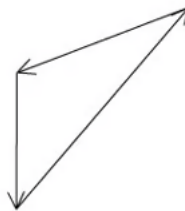
Question 10

A glider is descending at a constant speed at an angle of 15° to the horizontal. The diagram shows the directions of the lift **L**, air resistance **R** and weight **W** acting on the glider.



Which vector triangle could represent the forces acting on the glider?

A.



B.



C.

C



D.

D



[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- Combining $\Delta P = \frac{F}{A}$ and $F = ma$
 - $\Delta P = \frac{ma}{A} = \frac{kg \, ms^{-2}}{m^2} = kg \, m^{-1} s^{-2}$
- The units of density are
 - $\rho = \frac{m}{v} = kg \, m^{-3}$
 - $\frac{\Delta P}{\rho} = \frac{kg \, m^{-1} s^{-2}}{kg \, m^{-3}} = m^3 m^{-1} s^{-2} = m^2 s^{-2}$
 - $m \, s^{-1} = (m^2 s^{-2})^n$
- Therefore $n = \frac{1}{2}$

2

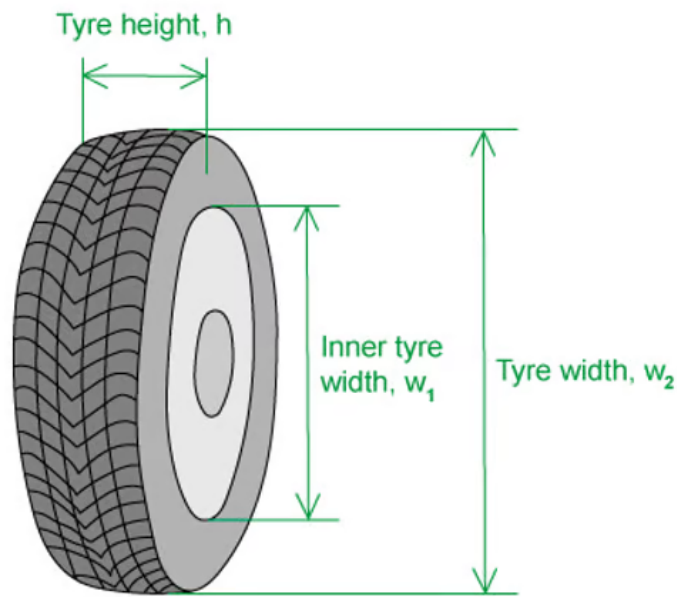
The correct answer is **D** because:

- Treating the tyre to be approximately cylindrical
 - $Volume = \pi r^2 h$, where h is the height of the tyre and r is half the width of the tyre, $r = \frac{1}{2}w$
- Volume of air in tyre = total volume of tyre - volume of inner tyre

$$= \pi h \left(\frac{1}{2}w_2 \right)^2 - \pi h \left(\frac{1}{2}w_1 \right)^2$$

$$= \frac{\pi h}{4} (w_2^2 - w_1^2)$$
- Some reasonable estimates are: $h \sim 15 \, \text{cm}$, $w_1 \sim 30 \, \text{cm}$ and $w_2 \sim 60 \, \text{cm}$

- Volume of air in tyre $\sim \frac{\pi \times 0.15}{4} (0.6^2 - 0.3^2) = 0.03 \text{ m}^3$



A is incorrect because a bus weighs at least several thousand tons and would be travelling at least 20 ms^{-1} meaning its kinetic energy would be in the region of $10^6 - 10^7$

B is incorrect because the standard wattage of a domestic light bulb is 60 W

C is incorrect because 300 K in Celsius is $300 - 273 = 27^\circ\text{C}$, whereas a hot oven would be at least 200°C which is $200 + 273 = 473 \text{ K}$

3

The correct answer is **C** because:

- The wavelength of UV light is about 10nm
- Using the wave equation $c = f\lambda$

- $f = \frac{3 \times 10^8}{10 \times 10^{-9}} = 3 \times 10^{16} \text{ Hz}$

- This is incorrect by a factor of 10^4

A is incorrect because statement is correct, 1 atm is equal to 1×10^5 Pa

B is incorrect because the time it takes for light to reach the Earth from the Sun is calculated by

$$\frac{1AU}{\text{Speed of light}} = \frac{1.5 \times 10^{11}}{3 \times 10^8} = 500s = 5 \times 10^2 s$$

D is incorrect because if we calculate the life-span of the person in the question we

get $\frac{2 \times 10^9}{365 \times 24 \times 60 \times 60} = 63$ years, which is a reasonable estimation

4

The correct answer is **B** because:

- Once any constants are removed, $F \propto \eta v$

$$\circ \quad \eta \propto \frac{F}{aV} = N m^{-1} (m s^{-1})^{-1}$$

$$= kg m s^{-2} m^{-2} s$$

$$= kg m^{-1} s^{-1}$$

- Therefore, the units for η are $kg m^{-1} s^{-1}$

5

The correct answer is **A** because:

- When $d = k_1 v^2$ the unit of k_1 is

$$\circ \quad \frac{d}{v^2} = [m][m s^{-1}] = m^{-1} s^2$$

- When $d = kv^2$ the unit of k is

$$\circ \frac{d}{v^2}(m)[kmh^{-1}] = m(ms^{-1})^{-2} \left(\frac{1000}{60 \times 60} \right)^{-2} = 13.0 m^{-1}s^2$$

- Comparing the two units we can see $k_I = 13.0 k$, so $k = 0.0772 k_1$

B and **C** are incorrect because these are the answers you get if you don't square the conversion i.e $\frac{3000}{1000} = 3.6$

D is incorrect answer because k_1 is equal to 13.0 k, not the other way around

6

The correct answer is **B** because:

- Since C_d is a quantity with no units we can disregard it for this question

- That leaves us with $F \propto \rho v^n A$

- So, $v^n \propto \frac{F}{\rho A}$

- $(ms^{-1})^n = [N][kg m^{-3}]^{-1}[m^2]^{-1}$

$$= kg m s^{-2} kg^{-1} m^3 m^{-2}$$

$$= m^2 s^{-2}$$

- Comparing $(m s^{-1})^n = m^2 s^{-2}$ we can see that $n = 2$

7

The correct answer is **B** because:

Method 1

- Using either the triangle or parallelogram method it's possible to determine the direction of the resulting vector
 - In both methods, you must slide the vectors, parallel to their current direction, so that they are arranged tip to tail
 - The resultant vector is the one now joining from tail to tip on the other

$$= kg \, m \, s^{-2} kg^{-1} m^3 m^{-2}$$

$$= m^2 s^{-2}$$

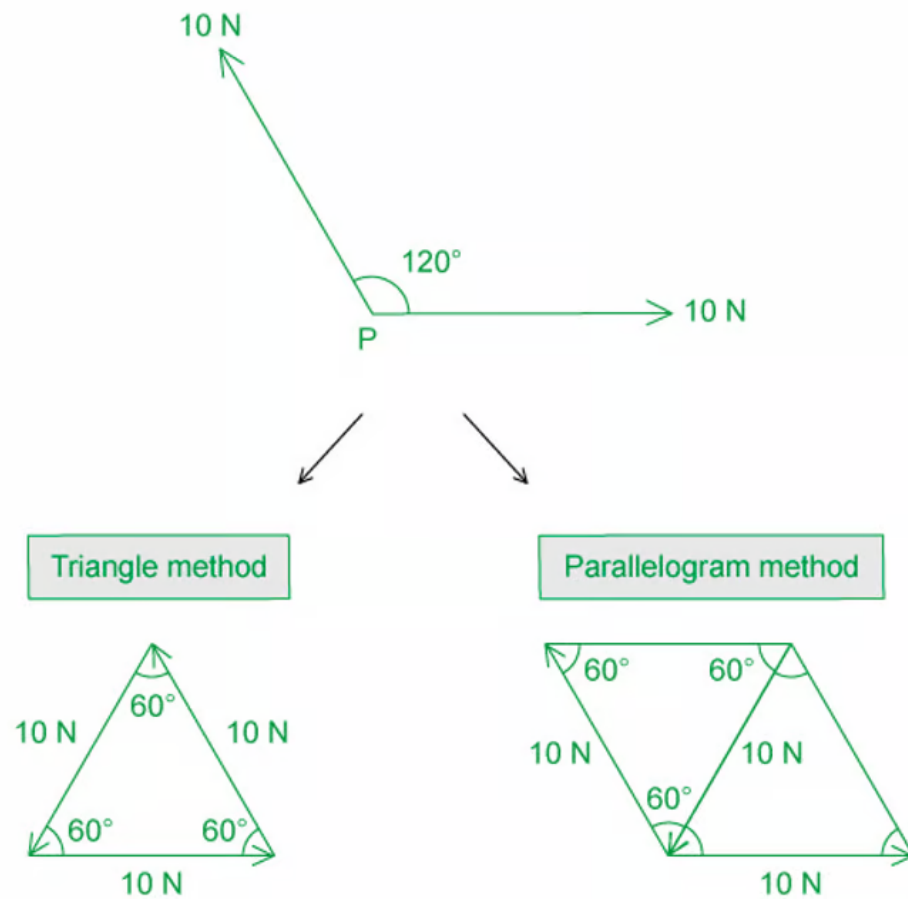
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7

The correct answer is **B** because:

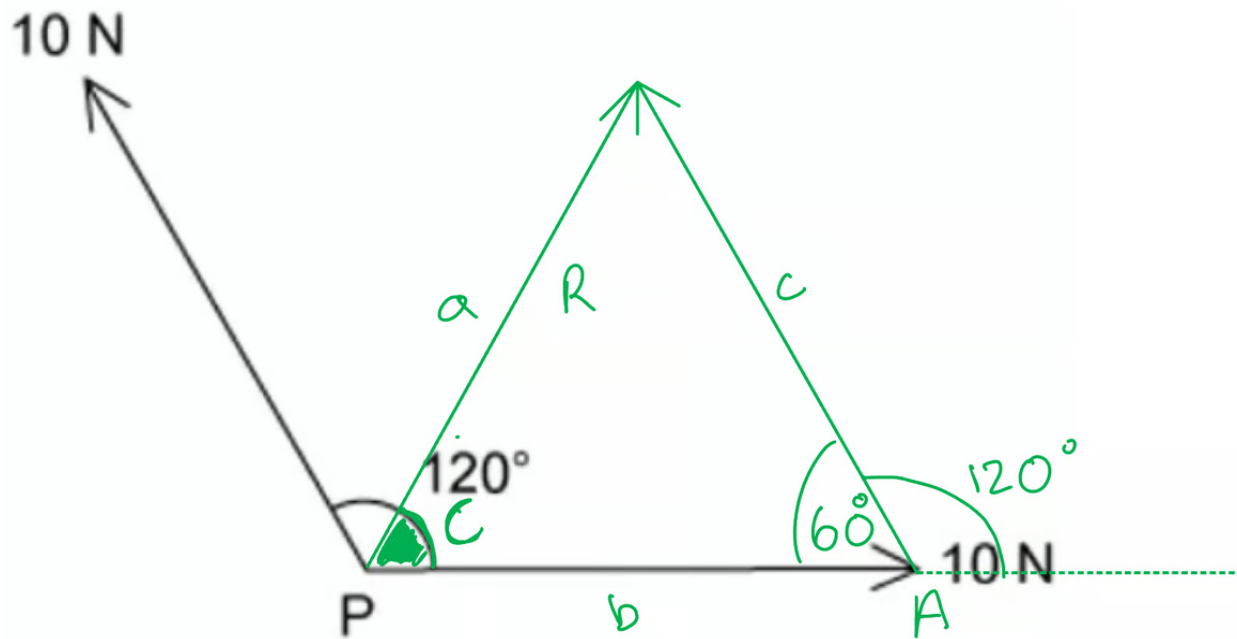
Method 1

- Using either the triangle or parallelogram method it's possible to determine the direction of the resulting vector
 - In both methods, you must slide the vectors, parallel to their current direction, so that they are arranged tip to tail
 - The resultant vector is the one now joining from tail to tip on the other side
 - The 120° angle will be on the outside of the triangle or parallelogram
 - So, the interior angle = $180 - 120 = 60^\circ$
- As shown in the diagram below, you can see that:
 - The vector triangle formed is an equilateral triangle
 - Where each angle is equal to 60° and the magnitude of each vector is 10 N
 - The resultant vector is the same length as the 10 N vectors, as the diagram is drawn to scale



Method 2

- Consider the cosine rule to find the length of the resultant side:
 - $a^2 = b^2 + c^2 - 2bc\cos(A)$
 - Label the angles and sides of the triangle



- Substitute in the known sides and angle
 - $a^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos(60)$
- Calculate the length of the resultant side a :
 - $a = 10 \text{ N}$
- Consider the sine rule to find the angle of the resultant side to the horizontal:
 - $\frac{a}{\sin(A)} = \frac{c}{\sin(C)}$
- Substitute in the known sides and angle:
 - $\frac{10}{\sin(60)} = \frac{10}{\sin(C)}$
- Rearrange the equation and calculate the angle of the resultant side to the horizontal:
 - $10 \times \sin(C) = 10 \times \sin(60)$
 - $\sin(C) = \sin(60)$
 - So, $C = 60^\circ$

8

The correct answer is **C** because:

- The unit of
 - v is m s^{-1}
 - g is m s^{-2}

- λ is m
- The unit of $g\lambda$ is
 - $(m s^{-2} m)^{\frac{1}{2}} = (m^2 s^{-2})^{\frac{1}{2}} = m s^{-1}$
- The units of the LHS and the RHS match so the correct answer is C

A is incorrect because the unit of $\sqrt{\rho g h}$ is $([kg m^{-3}][m s^{-2}][m])^{\frac{1}{2}} = (kg m^{-1} s^{-2})^{\frac{1}{2}}$

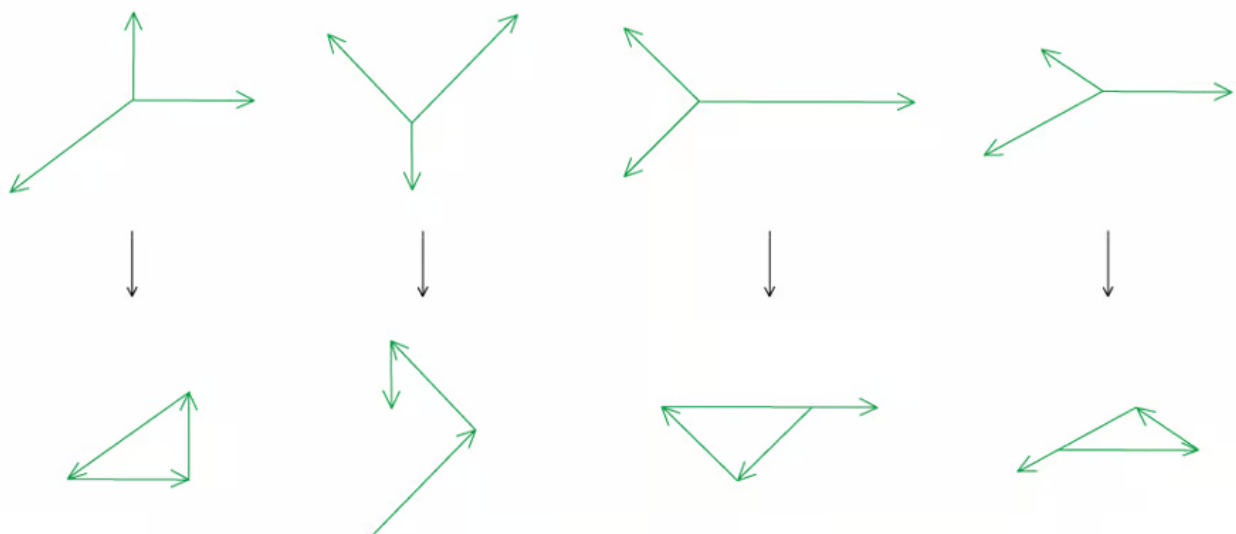
B is incorrect because the unit of $\sqrt{\frac{g}{h}}$ is $([m s^{-2}][m^{-1}])^{\frac{1}{2}} = s^{-1}$

D is incorrect because the unit of $\sqrt{\frac{g}{\rho}}$ is $([m s^{-2}][kg^{-1} m^3])^{\frac{1}{2}} = (kg^{-1} m^4 s^{-2})^{\frac{1}{2}}$

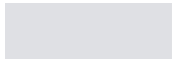
9

The correct answer is **A** because:

- Equilibrium means the forces are balanced. For 3 forces in equilibrium, the force vectors always form a closed triangle
- Constructing vector triangles (to scale, therefore, it is recommended you use a ruler), we can see the only balanced system is A
- This is because the sizes and directions of the forces align and form a closed triangle, satisfying the condition that the net force must be zero

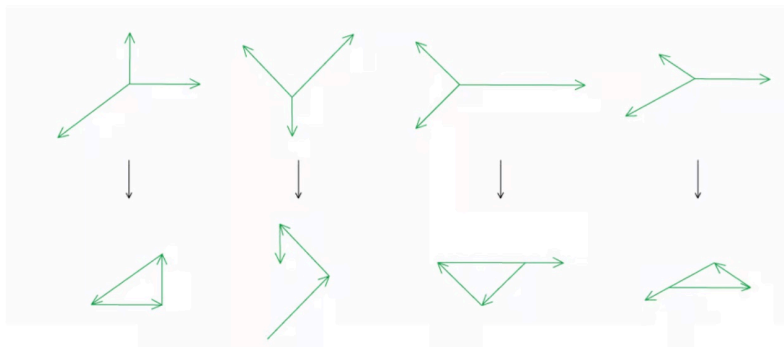


B, C & D are incorrect because none of these systems of coplanar forces form a closed triangle meaning the net force is not zero



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B, C & D are incorrect because none of these systems of coplanar forces form a closed triangle meaning the net force is not zero

The correct answer is **D** because:

- When you examine the individual vectors in the scenario you can immediately whittle the selection down to either B or D as both correctly represent vectors **L** and **W**
- On further inspection, you can see vector **R** is inclined 15° to the horizontal in a north-west direction so the correct answer is D

